Exploratory Data Analysis (EDA): One Categorical/Numerical Variable

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MATH 240

Lecture #5 — Ch. 4.4, 4.5, 5.2-5.6

Reminders

- Homework #1 is due **tonight** at 11:59pm
  - Includes textbook problems and R problem
- Checkpoint #1 is **today** towards the end of class

Learning Objectives

- **Analyze** metrics of a single numerical variable using five-number summaries
- **Visualize** a single numerical variable using histograms and box plots

Movies Dataset

Today, we'll work with data on movies as a way for us to explore various types of data.

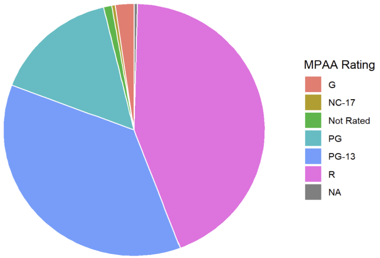
- 3,039 movies made in U.S. between 1929 and 2016, scraped from IMBD (by Dr. Robin Belton)

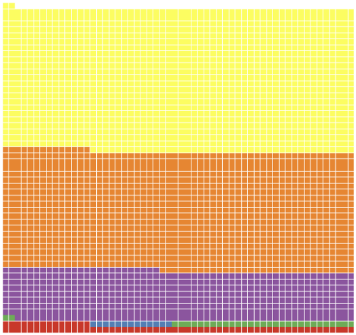
	movie_title	title_year	budget	log_budget	gross	log_gross
1	Avatar	2009	237000000	19.28357	760505847	20.44949
2	Pirates of the Caribbean: At World's End	2007	300000000	19.51929	309404152	19.55016
3	The Dark Knight Rises	2012	250000000	19.33697	448130642	19.92060
4	John Carter	2012	263700000	19.39032	73058679	18.10677
5	Spider-Man 3	2007	258000000	19.36847	336530303	19.63420
6	Tangled	2010	260000000	19.37619	200807262	19.11786

Figure: First Six Movies in Movies Dataset

One Categorical Variable: Pie and Waffle Charts

- Use of continuous area to show relative size of categories
  - e.g. pie charts, stacked bar plot
- On a discrete level: waffle chart





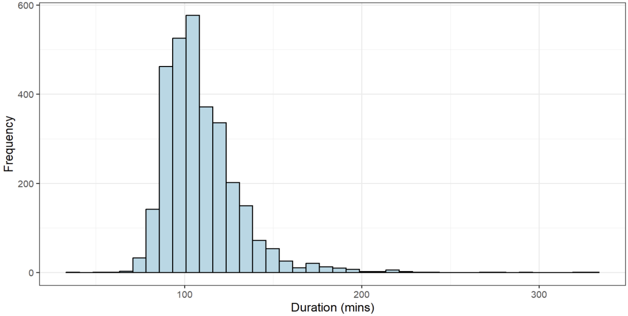
The Factor Function

- `factor()` function is useful for associating categories/levels with a variable

```
factor(movies$content_rating, ordered = F,
       levels = unique(movies$content_rating))
```

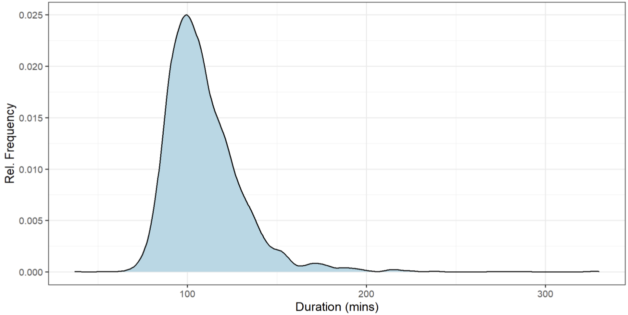
One Numerical Variable: Histogram

- Visualizes *distribution* of one **numeric** variable
- Three traits we note:
  - Center: where is the middle of our data?
    - e.g. median, mean
  - Spread: how spread out are our values?
    - e.g. standard deviation, range
  - Shape: where is the bulk of the data? what about outliers?
    - e.g. mode



One Numerical Variable: Density Plots

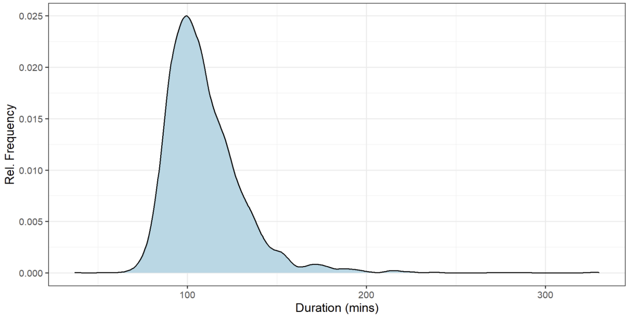
- We can visualize similar traits with a density plot
  - Use of standardization and smoothing over corresponding histogram



One Numerical Variable: Skewness

Skewness measures how symmetric (or asymmetric) our distribution is

- Use of histograms, density plots (can use box plots too!)



Skewness will dictate which measurements we use for central tendency and spread.

One Numerical Variable: Measures of Central Tendency

Central tendency measures the center of our data.

- **Mean**: average of data (sum up all data values, divide by number of values)
  - $\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$
- **Median**: 50th percentile of data (half of the data points fall below this value, half are above)
  - $x_{\frac{n+1}{2}}$ if n is odd
  - $x_{\frac{n}{2}} + x_{\frac{n}{2}+1}$ if n is even

One Numerical Variable: Measures of Spread

Spread measures how far our data extends from the center.

Standard deviation:

$$
\begin{aligned}
s &= \sqrt{s^2} \\
&= \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}
\end{aligned}
$$

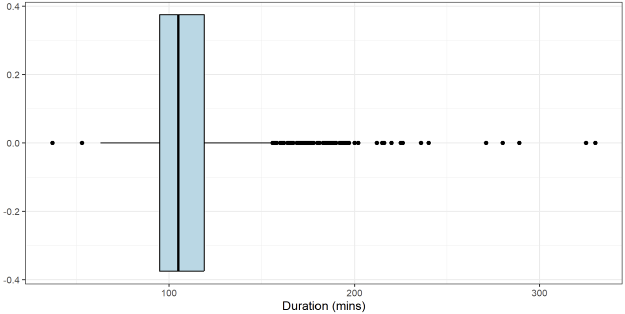
Interquartile range:

$$IQR = Q_3 - Q_1$$

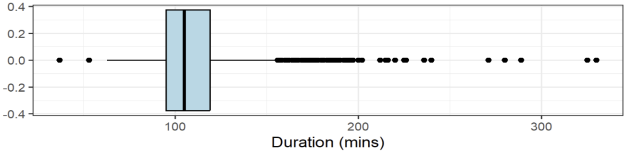
- The difference between the 75th and 25th percentiles (i.e. 3rd and 1st quartiles)

One Numerical Variable: Box Plot

A **box plot** visualizes the quartiles of our distribution, as well as any outliers:



One Numerical Variable: Five-Number Summary



A five-number summary captures information about the center and spread of data:

- Minimum: Lowest value of our data
- 1Q: 25th percentile
- Median: 50th percentile
- Mean: average
- 3Q: 75th percentile
- Maximum: Highest value of our data

Checkpoint

- You are allowed to use your notes and R to complete this assignment. Any use of AI models, search engines, communication platforms, and someone else's notes are prohibited.
- Follow all directions and show all work as stated. If numerical answers are asked, round any decimal answers to four decimal places.
- When finished, turn in the assignment to me at the front of the classroom.
- You have until the end of the class period to finish.

For Next Week!

- Homework #1 is due **tonight** at 11:59pm
  - Includes textbook problems and R problem
- Homework #2 will be assigned over the weekend, due next Friday at 11:59pm
- Read chapters 4-5 from textbook before Monday's class